

Tejas Joshi

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Summary

I am an engineering enthusiast, always eager to learn, build and contribute towards development and innovation.

Education

College of Engineering Pune

B.Tech · Computer Engineering · India

Sep 2022 - Present

8.46 / 10.00

- Member of CoEP Satellite Team, working on building a solar sailing satellite
- Participant in Punt Formation and punt races in annual Regatta
- Organiser of technical event in annual technical festival MindSpark

Projects

Ext2 file system

· C, cscope, ctags

Mar 2025 - May 2025

Operating Systems

- Worked on xv6 codebase to modify relevant file system related system calls in C programming language
- Simulated a mount system call by writing relevant assembly code
- Involved adding a VFS layer to support ext2 file system along with existing xv6 file system

Digital Receipt Management System | [github repository link](#)

College of Engineering Pune · MERN, Tesseract OCR, AWS S3

Mar 2025 - May 2025

Software Engineering

- Created an all in one solution for managing receipts, recurring bills and expenses
- Provided functionality to add receipt data manually, fetch receipts from gmail or extract text from images using OCR
- Used AWS s3 service for long term storage of receipt images
- Practiced standard software development practices and methodical documentation process

Home Server

· Debian Server, Nextcloud, Bookstack

Jun 2025 - Present

Computer Networks

- Repurposed old hardware to manage personal data requirements such as photo storage, music storage
- Used Nextcloud service for collaborative data sharing
- Used bookstack to store vast collection of books
- Planning to host a technical blog on the server

Bash-like-shell

College of Engineering Pune · C/C++, Linux, gdb, cscope, ctags

Mar 2025 - Apr 2025

Systems Programming

- Implemented a simple bash like program for executing commands in linux environment
- Detailed use of linux system calls, along with C language
- Added functionality for I/O redirection and pipes

Gesture Controlled Robot

· C++, Arduino

May 2025 - Jul 2025

Embedded Systems

- A basic gesture controlled robot created using arduino, accelerometer, transmitter receiver and motor along with motor driver
- Explored protocols like I2C, SPI for interfacing various sensors

VoIP application

· Python

Nov 2024 - Dec 2025

Computer Networks

- Implemented a basic Voice over Internet Protocol application for voice communication within a network
- Discovered the complexity of VoIP, socket programming and softphones

Skills

Skills: C++, C Programming, Python, MySQL, Linux, Image Processing